

Data Ecosystems and Governance in Organizations

Msc in Business Analytics

Team 13

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Agenda

01

Data Quality Assessment

Identifying & remediating issues in the raw dataset

02

Bias and Fairness Analysis

Disparate Impact, age-based patterns & proxy discrimination

03

Privacy and Governance

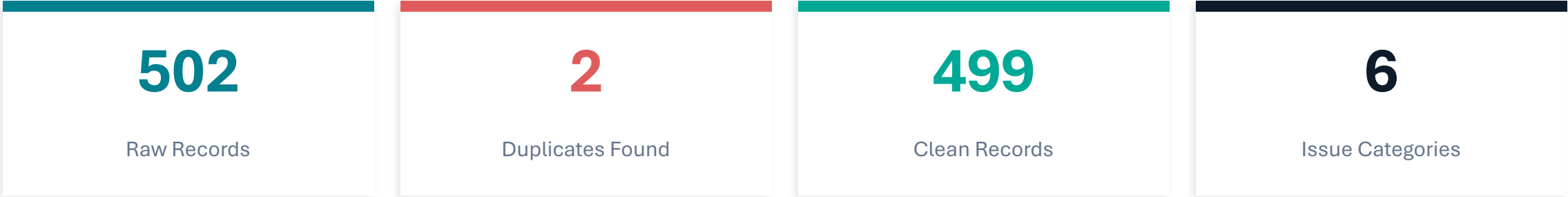
Sensitive data mapping, pseudonymisation & GDPR compliance

04

Governance Recommendations

Controls, audit trails & AI Act alignment

01 Data Quality Assessment

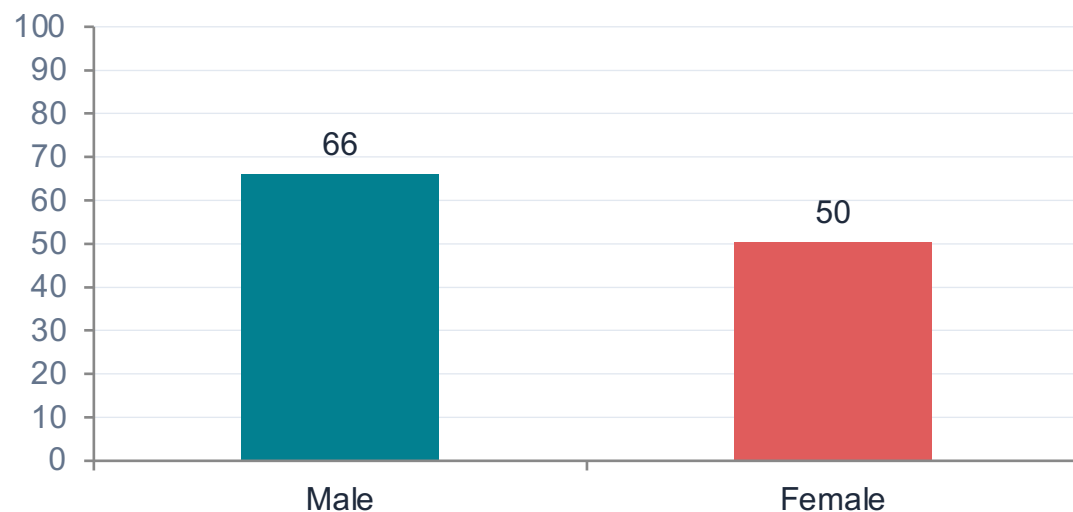


Inconsistent Date Formats	161 records (32.3%) — mixed ISO, DD/MM/YYYY, YYYY/MM/DD	Consistency
Gender Coding Abbreviations	114 records (22.7%) — M/F vs Male/Female + 2 empty values set to Unknown	Consistency
Income Data Type Mismatch	financials.annual_income: 3 types found (float, int, str) — standardised to float	Validity
Missing Values	email (7), DOB (4), IP (4), SSN (4), ZIP (1) across 500 records	Completeness
Duplicate Records	app_001 and app_042 duplicated — flagged with DUPLICATE_ENTRY_ERROR and removed	Uniqueness
Negative / Invalid Values	credit_history < 0: app_043 and app_156 corrected to 0. app_290 savings -5000 removed	Validity

02 Bias and Fairness Analysis

Disparate Impact Ratio

Four-Fifths Rule (DI < 0.8 = potential discrimination)



DI = 0.76

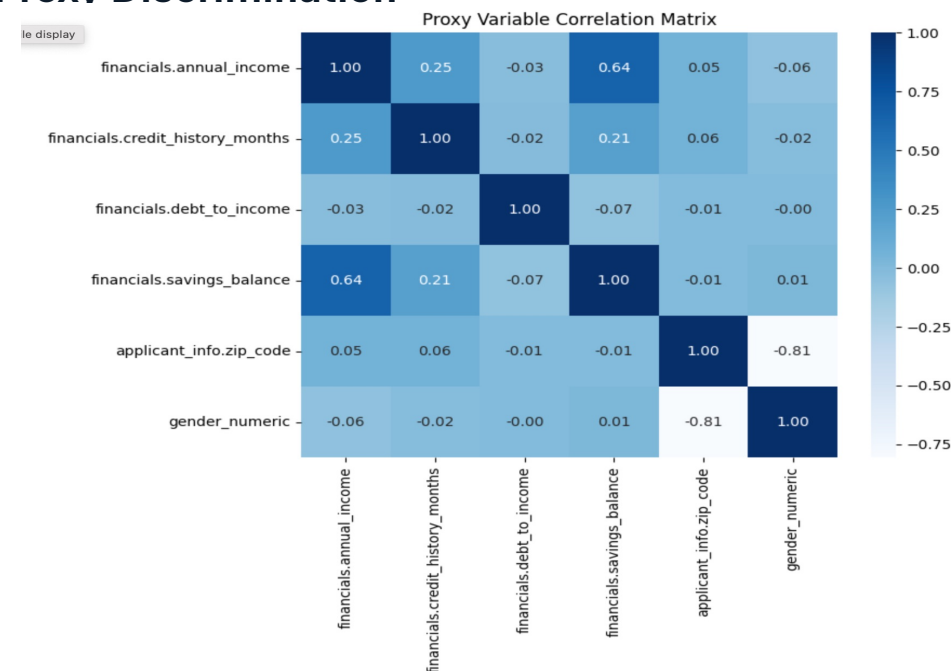
Below 0.80 threshold

Female 50.4% vs Male 66.0% → Both fail the 80% rule

Age-Based Patterns

Applicants aged 18–25 show 41.7% approval vs. 65.3% for 35–44 age group. DI = 0.556 for 18–24 — below 0.80 threshold.

Proxy Discrimination



03 Privacy and Governance

PII Fields Identified

full_name	HIGH	Art. 5 – Minimisation
email	HIGH	Art. 5 – Minimisation
ssn	CRITICAL	Art. 9 – Special category
ip_address	HIGH	Art. 4 – Personal data
date_of_birth	HIGH	Art. 9 – Special category
zip_code	MED	Art. 4 – Personal data

Pseudonymisation Demo

SHA-256 hashing applied to full name, email, and SSN. IP address and spending behavior columns fully removed (data minimisation).
Personal Identifiable Information classified into 3 tiers: direct identifiers, indirect/special identifiers, and financial personal data.

GDPR & EU AI Act Mapping

- Art. 5 — SHA-256 pseudonymisation: full name, email, SSN
- Art. 6 — No documented lawful basis for data collection
- Art. 17 — No right-to-erasure mechanism in place
- Art. 22 — Automated decisions: 80%+ opaque algorithm risk score rejections

04 Governance Recommendations

1 Data Validation Pipeline

Enforce schema types, date format standards, and categorical value constraints at ingestion. Automated anomaly alerts for future loads.

2 Fairness Monitoring

Compute Disparate Impact ratio on every model retrain. Trigger manual review when Disparate Impact < 0.8 . Monitor intersectional effects (gender $\times$ age).

3 Proxy Attribute Audit

Remove or control ZIP code from model features. Evaluate all geographic/spending variables for protected-characteristic correlation before training.

4 Privacy-by-Design

Pseudonymise Personal Identifiable Information at rest. Implement consent tracking, data retention schedules and GDPR Art. 17 erasure workflows.

5 Audit Trail & Explainability

Log all automated decisions with model version, input features, and outcome. Require human review escalation for borderline cases (EU AI Act Art. 14).

6 Model Governance Board

Quarterly fairness report to senior management. Documented approval process before any credit model goes live. Role-based data access control.

Key Metrics

Data Quality

5 issue categories identified and remediated —> dataset reduced to 500 clean, validated records.

Bias Detected

Disparate Impact ratio of 0.78 (< 0.80) signals potential gender disparate impact requiring immediate model review.

Privacy Gaps

6 Personal Identifiable Information fields found unprotected; Social Security Numbers and IPs pose critical GDPR and AI Act compliance risk.

Governance

6 actionable controls proposed, from validation pipelines to fairness monitoring and human oversight.